

Chronic Kidney Disease Prediction Using Machine Learning Algorithms and the Important Attributes for the Detection

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Abstract - In the modern world, chronic kidney disease has become one of the most hazardous diseases. CKD is a condition in which the kidney cannot perform the proper filtering of the blood or it stopped working completely which causes the left toxic into the blood, which leads the patient to death. It is likely impossible to detect CKD in the early stages, and it is very difficult to save patient's lives in the last stage of CKD. A patient's life can be saved by renal transplant or the early detection the CKD. Machine Learning algorithm techniques have played a very important role in CKD prediction. Past medical test, reports can also be used as a tool for the early detection of renal disease. Machine Learning (ML) Techniques like KNN, Decision Tree, and ANN are used in this review. We have to find out that Decision Tree has shown the best result of 98.60% of accuracy. Generally, the majority of the algorithms are based on supervised learning and classification problem solving. We have explained some important attributes, which play a major role in early CKD Prediction and Detection. Every attribute has its specific effect on CKD. The previous researchers have done many experiments to get the best attribute and best ml technique for the prediction. in this paper, we have studied all related ML techniques, and important Attributes and discussed the measurement factor for CKD prediction.

Keywords: CKD, Renal Disease, Machine Learning, Neural network, GFR, Diabetes Mellitus, Cardio Vascular disease (CVD), Creatinine.

I. INTRODUCTION

Chronic kidney disease is a life-threatening condition in which the kidney is not able to filter the Blood or it is totally failed to perform any task, then the toxic in the blood remains in the blood, which kills the patient slowly, Inability of any Prior test can detect the CKD in its early stages. The only tests for CKD are the GFR test which is very expensive but not able to detect the early stages of the CKD, and these tests are not so common they take some specific test labs, which are not available easily to very patients. The CKD is Generally divided into the 5 stages based on the glomerular filtration rate (GFR) [figure 1] [8]. In the First stage of CKD, kidneys do some insignificant errors in blood filtering. In the second stage of the CKD, the error is increased a little bit which the patient cannot detect. In the third stage of CKD, the error in the blood filtering is more increased but the

patient also cannot detect it. In the Fourth stage of the CKD, the error is significantly increased and the patient starts to have some serious symptoms of kidney-related issues, and if the patient ignores this problem that leads to the Fifth stage of the CKD in which the kidney stop working completely. The only possible way to save the patient's life in the fifth stage is the renal transplant, which is very difficult as well as expensive. Hence, for saving a patient's life from CKD is to somehow to detect in its early stages. Where the nephrologist can give the medicine to stop the infection in the kidney. We can also use the past medical records for the detection of the CKD. The past medical record consists a lot of information about the patient like age, blood pressure, sugar, hemoglobin, anemia and etc. these attributes can also be very useful for CKD detection because the human's organs are interconnected to each other and perform the task with help of others. Like we can say that blood is pumped by the heart through all over the body and the kidney filters the blood and eliminates the toxic form it, hence if there is any problem with the heart the blood flow will be changed and that will affect the kidney filtering. Smoking and depression can also affect kidney filtering. Hence, we can use all the past records for the future detection of the CKD [3].

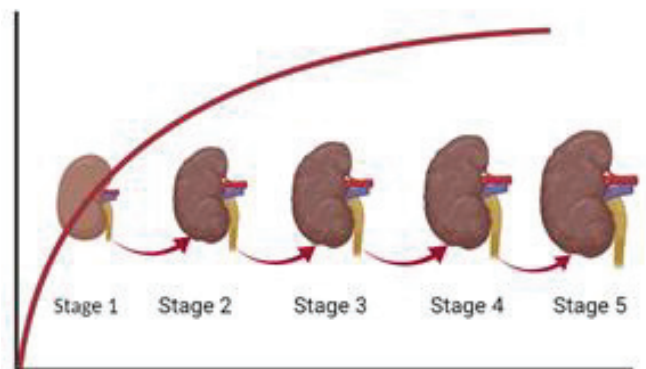


Fig. 1. Stages and infection of the CKD

II. LITERATURE REVIEW

CKD is a one of the most sever disease, which is creating so many life losses. More than 10% of people are facing this

critical renal problem. For the future prediction there are many researchers have done on the CKD using Machine Learning Techniques [13], previously so many of the ML algorithms were used for the prediction of CKD. ML Techniques used for the CKD prediction and detection are listed below:

A. K-Nearest Neighbor

It is known as the KNN technique. It is a classification Algorithm and part of supervised learning. It is used for knowledge Extraction such as classification and regression problems. Many researchers for classification work use KNN. Bashir et al. [1] created a framework using multi-layer classification for the detection of diseases. After implying KNN they got 57.41% of accuracy using a dataset from startling and while using a dataset of Cleveland they got 58.42% of accuracy. Hasi et al. [2] have implied KNN on Pima Indians dataset for CKD their research shown KNN has generated 76.96% of accuracy with very less error rate for Chronic Kidney Disease detection. **The Pima Indians dataset has 768 samples, which include 8 numerical value attributes.** Khan et al. Used KNN in UCI Machine Learning Repository for Lever infection detection, their result of KNN produced an accuracy of 62.90% and the error rate of 0.3718.

B. Decision Tree

A decision tree (DT) is a supervised learning method that may be applied as both a classification and regression tool. Generally, it is structured as a tree classifier where internal nodes consider as data set features, branches serve as decision rules and each leaf node presents the result. Sharma et al. [4] have used a decision tree for the Chronic Kidney Disease prediction and got 98.60 % of accuracy, specificity of 1, and precision rate of 1, by using the UCI Machine Learning Repository dataset. Asker et al. [5] used Decision Tree for renal disease prediction using 24 features from the UCI ML Repository data set. They got 98.4127% of accuracy and an error rate of 0.0159 [15].

C. Artificial Neural Network

Artificial Neural Network is a sub-type of artificial intelligence and works as a supervised learning method. Artificial Neural Network has the neurons like a human brain, and every neuron is connected with other neurons in-network and these neurons are known as nodes. Artificial Neural networks have 3 layers. The input layer is the top layer, the concealed layer is the next, and the output layer is the last. The input layer retrieves input from the dataset that is sent to Hidden Layer. Hidden Layer processed the input data and find the hidden pattern from this it is sent to the Output Layer where the result is generated. Chithra et al. [6] used the ANN for the CKD prediction over **TCIA dataset** where they got 95.20% of accuracy and after using UCI ML Repository data set they got 91.92% of accuracy while taking full features in the dataset. Vijayrani et al. [7] used ANN for Renal disease prediction. **They collected the data from different labs, hospitals, and medical centers. They achieved 87.70% accuracy in their experiments.**

1) Age

The age is a numerical value, which is measured in the form of the year. In previous studies we have found that people of age group 45+ have more chances of suffering from renal Disease. Age is a very important attribute in the case of CKD [12]. So on the basis of early studies, we can

say that high aged patients have more chances of CKD infection [9]. (age distribution chart figure 2)

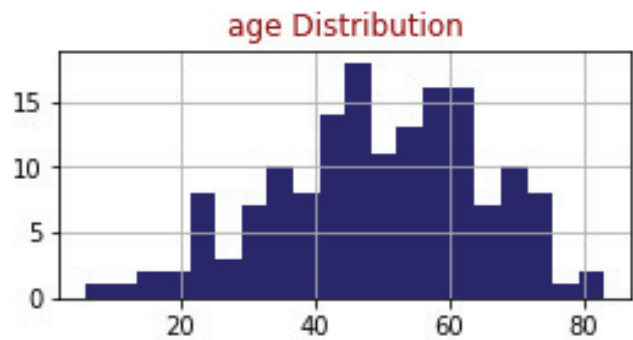


Fig. 2. Age Distribution Chart

2) Blood Pressure

BP is measured in numerical values. The Measurement unit of the Blood Pressure is millimeters of mercury (mmHg) in our blood vessels. Blood Pressure is can be defined as the pressure that is generated by the Blood flow in or blood veins. In general, if BP is increased in a patient's body that can trigger the CKD or increase the chances of CKD occurrence [10] [14]. (Blood Pressure distribution chart figure 3).

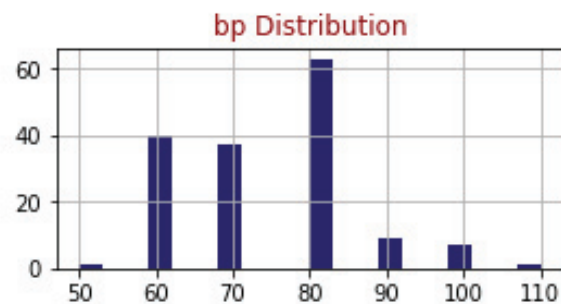


Fig. 3. Blood Pressure Distribution Chart

3) Specific Gravity

Specific Gravity can be used for the measurement of the kidney's capacity to dilute or concentrate urine into plasma because urine is a solution of minerals, Salts, and compounds dissolved in water. The urine is more concentrated the higher the specific gravity. **If the specific gravity is less it means less concentrated urine.** Hence kidneys need to work harder for the removal of the toxins. Then there are more chances for renal damage or CKD [11]. (sg distribution chart figure 4)

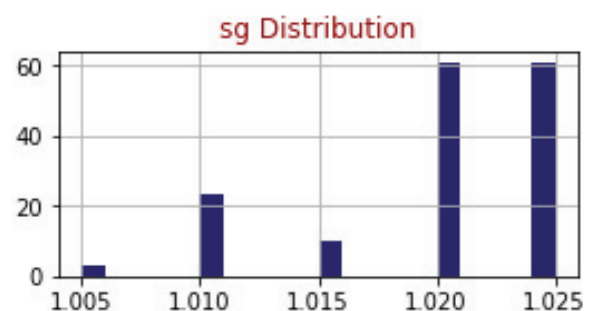


Fig. 4. Specific Gravity Distribution Chart

III. DISCUSSION

We have so many of ML and AI techniques, which are used for the early detection, and prediction of the renal disease. In addition, the past records can help to get the diseased outcome. Hence, we have some Measurement method for calculating the results we got after performing the techniques on the attributes. The Measurement Methods are listed below-

A. Mean Absolute Error:

MAE is an amount to calculate the prediction, which deviate from the true probability. $P(j,k)$ is the approximate of coefficient to be class k . MAE takes value between 0 to 1, and is given by-

$$MAE = \frac{\sum_{a=1}^n |y_a - x_a|}{n}$$

MAE = Mean absolute error

y_a = Prediction

x_a = True value

N = total number of data points

B. Root Mean Squared Error

RMSE is the square root of the mean span between predicted and remarked data. Signifying the approximated probability of the acceptable data j and the selected value for the acceptable data k by $p(j,k)$ and T_k independently, RMSE is calculated as follow-

$$RMSE = \sqrt{\frac{\sum_{i=0}^Z \hat{a}_i - a_i}{Z}}$$

RMSE = root mean square error

I = variable I

$\hat{a}_i$ = estimated time series

a_i = actual observations time series

Z = number of non missing data points

C. Relative Absolute Error

RAE is the way to help in prediction of performance of different models. This is used in different area like machine learning, data mining, etc. Using below formula, it can be calculated-

$$\delta = \left| \frac{U_a - U_e}{U_e} \right|$$

δ = percent error

U_a = actual value note

U_b = expected value

D. Root Relative Squared Error (RRSE)

RRSE is used for basic level predicted. It takes total squared error

$$RRSE_i = \sqrt{\frac{\sum_{j=1}^n (U_{(ij)} - V_j)^2}{\sum_{j=1}^n (V_j - \bar{V})^2}}$$

$U_{(ij)}$ = Predicted value by model

V_j = Target value

$$\bar{V} = \frac{1}{n} \sum_{j=1}^n V_j$$

TP = number of true positives

TN = number of true negatives

FP = number of false positives

FN = number of false negatives

E. Recall

It is known as the average probability of total retrieval, and calculated by-

$$Recall = \frac{TP}{TP + FN}$$

F. Precision

It is known as related number between the accurately classified module to whole classified error-prone modules. Precision is the number of accurately prescribed as error:

$$Precision = \frac{TP}{TP + FP}$$

G. F-Measure

It is one of union of both Recall and Precision. F-Measure uses the retrieval information to approximate classification performance and is given by:

$$F\text{-Measure} = \frac{TP}{TP + \frac{1}{2}(FP + FN)}$$

H. Accuracy

It is given by the percentage of accurately classified cases, as well as, accuracy is one of the most useful classification matrix in classification data and calculated by-

$$Accuracy = \frac{TN + TP}{FP + TP + TN + FN}$$

Type of relation and inter-relation between themselves and CKD [figure 2]. An age distribution Chart [figure 2] is also created by the tableau for the better understanding of the dataset and age effect in the CKD. We gave generated the distribution chart for the every single attribute in figure 31.

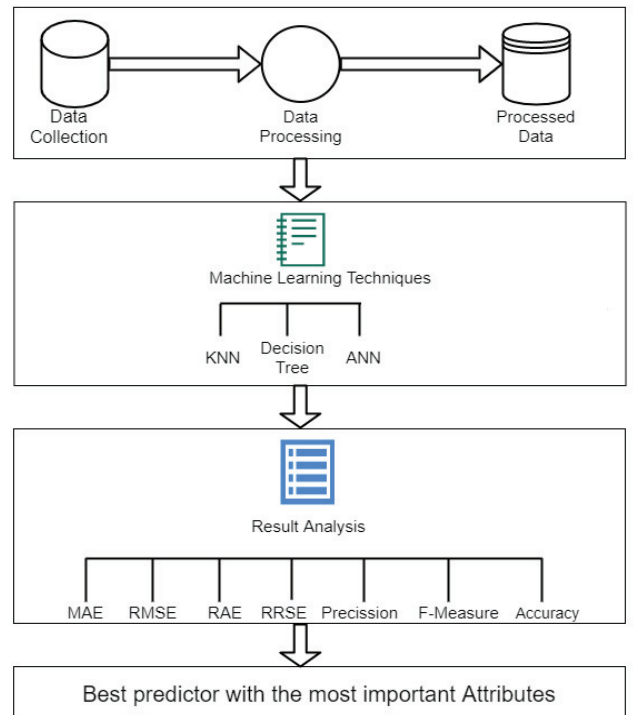


Fig. 5. Data Flow Diagram

TABLE I. ACCURACY TABLE

S No.	ML Techniques	Accuracy(%)
1	KNN	76.96
2	Decision Tree	98.60
3	ANN	95.20

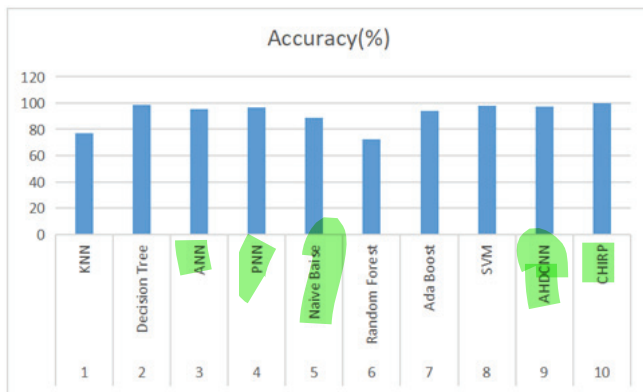


Fig. 6. Accuracy Graph

TABLE II. MOST IMPORTANT ATTRIBUTES

S.No	Attribute	Description	Class	Measure Scale
1	sg	Specific Gravity	Nominal	1.005,1.010,1.015,1.020,1.025

TABLE III. AVERAGE IMPORTANT ATTRIBUTE

S.No	Attribute	Description	Class	Measure scale
1	bp	Blood Pressure	Numerical	In mm/Hg

TABLE IV. LEAST IMPORTANT ATTRIBUTE

S.No.	Attribute	Description	Class	Measure Scale
1	age	Age in year	Numerical	Age in Year

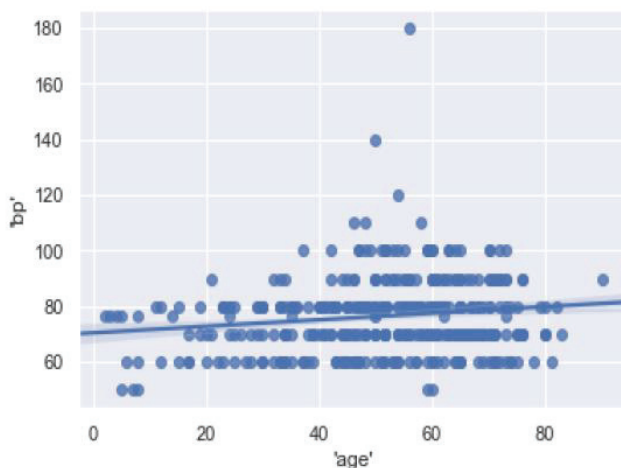


Fig. 7. Relation between bp and age

IV. CONCLUSION

Recent study have shown that CKD is one of the most sever disease. Inability of the early detection tests are the reason for the high death rate in CKD. Nephrologist have

very few test for the CKD detection, which are not so much available in every region and are very much expensive too. In the last stage of the CKD renal transplant in the only way save patient life and the surgery success rate is very low. Hence, the early detection of the renal ways is the best cure for this life-threatening problem. ML and techniques are the best possible solution for this problem, which is also developing day by day. We can also use the past medical reports for the CKD detection, they include many important attributes which are used by the nephrologist and researchers. Knowledge of these attribute can give a lot of idea about the kidney condition and could be used as the early detection method for the CKD.

REFERENCES

- [1] Bashir, S., Qamar, U., Khan, F. H., & Naseem, L. (2016). HNV: a medical decision support framework using multi-layer classifiers for disease prediction. *Journal of Computational Science*, 13, 10-25.
- [2] Hashi, E. K., Zaman, M. S. U., & Hasan, M. R. (2017, February). An expert clinical decision support system to predict disease using classification techniques. In *2017 International Conference on Electrical, Computer and Communication Engineering (ECCE)* (pp. 396-400). IEEE.
- [3] Khan, B., Naseem, R., Ali, M., Arshad, M., & Jan, N. (2019). Machine learning approaches for liver disease diagnosing. *International Journal of Data Science and Advanced Analytics* (ISSN 2563-4429), 1(1), 27-31.
- [4] Sharma, S., Sharma, V., & Sharma, A. (2016). Performance based evaluation of various machine learning classification techniques for chronic kidney disease diagnosis. *arXiv preprint arXiv:1606.09581*.
- [5] Alasker, H., Alharkan, S., Alharkan, W., Zaki, A., & Riza, L. S. (2017, October). Detection of kidney disease using various intelligent classifiers. In *2017 3rd international conference on science in information technology (ICSITech)* (pp. 681-684). IEEE.
- [6] Chittora, P., Chaurasia, S., Chakrabarti, P., Kumawat, G., Chakrabarti, T., Leonowicz, Z., & Bolshev, V. (2021). Prediction of Chronic Kidney Disease-A Machine Learning Perspective. *IEEE Access*, 9, 17312-17334.
- [7] Vijayarani, S., Dhayanand, S., & Phil, M. (2015). Kidney disease prediction using SVM and ANN algorithms. *International Journal of Computing and Business Research (IJCBR)*, 6(2), 1-12.
- [8] Khan, B., Naseem, R., Muhammad, F., Abbas, G., & Kim, S. (2020). An empirical evaluation of machine learning techniques for chronic kidney disease prophecy. *IEEE Access*, 8, 55012-55022.
- [9] Prakash, S., & O'Hare, A. M. (2009, September). Interaction of aging and chronic kidney disease. In *Seminars in nephrology* (Vol. 29, No. 5, pp. 497-503). WB Saunders.
- [10] Constantiner, M., Sehgal, A. R., Humbert, L., Constantiner, D., Arce, L., Sedor, J. R., & Schelling, J. R. (2005). A dipstick protein and specific gravity algorithm accurately predicts pathological proteinuria. *American journal of kidney diseases*, 45(5), 833-841.
- [11] Menon, V., Wang, X., Greene, T., Beck, G. J., Kusek, J. W., Marcovina, S. M., & Sarnak, M. J. (2003). Relationship between C-reactive protein, albumin, and cardiovascular disease in patients with chronic kidney disease. *American Journal of Kidney Diseases*, 42(1), 44-52.
- [12] Navaneethan, S. D., Schold, J. D., Jolly, S. E., Arrigain, S., Winkelmayer, W. C., & Nally Jr, J. V. (2017). Diabetes control and the risks of ESRD and mortality in patients with CKD. *American Journal of Kidney Diseases*, 70(2), 191-198.
- [13] Malhotra, R., Craven, T., Ambrosius, W. T., Killeen, A. A., Haley, W. E., Cheung, A. K., & SPRINT Research Group. (2019). Effects of intensive blood pressure lowering on kidney tubule injury in CKD: a longitudinal subgroup analysis in SPRINT. *American Journal of Kidney Diseases*, 73(1), 21-30.
- [14] Umanath, K., & Lewis, J. B. (2018). Update on diabetic nephropathy: core curriculum 2018. *American Journal of Kidney Diseases*, 71(6), 884-895.
- [15] Sobrinho, A., Queiroz, A. C. D. S., Da Silva, L. D., Costa, E. D. B., Pinheiro, M. E., & Perkusich, A. (2020). Computer-aided diagnosis of chronic kidney disease in developing countries: A comparative analysis of machine learning techniques. *IEEE Access*, 8, 25407-25419.